

Template Title Homework

Setting problem as a section

Problem 1

This is a problem

(a)

This is a subsection of this problem

Solution :

We can also make problem as a theorem

Problem 1 *this is a problem*

Exercise 2 *this is a exercise*

1. $T(n) = 2T(\lfloor n/4 \rfloor) + n^{1/2}$.
2. $T(n) = 3T(\lfloor n/2 \rfloor) + n \lg n$.
3. $T(n) = 5T(\lfloor n/5 \rfloor) + \frac{n}{\lg n}$.
4. $T(n) = 4T(\lfloor n/2 \rfloor) + n^2 \sqrt{n}$.
5. $T(n) = 2T(\lfloor n/2 \rfloor) + n \lg n$.

$a = 3, b = 2$ implies a reference function $g(n) = n^{\log_2 3}$. Converting as follows,

$$\begin{aligned} y &= \log_2 3 \\ 2^y &= 3 \\ y \ln 2 &= \ln 3 \\ y &= \frac{\ln 3}{\ln 2} = 1.585, \end{aligned}$$

we have $g(n) = n^{1.585}$. The “glue” function is $f(n) = n \lg n$. Let $g_\epsilon(n) = n^{1.585-\epsilon}$, for $0 < \epsilon < 0.5$. Since

$$\begin{aligned} \frac{f(n)}{g_\epsilon(n)} &= \frac{n \lg n}{n^{1.585-\epsilon}} = \frac{\lg n}{n^{0.585-\epsilon}} \\ &\leq \frac{\lg n}{n^{0.085}} \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$, we have $f(n) = o(g_\epsilon(n))$, which implies $f(n) = O(g_\epsilon(n))$ and allows case (1) of the master template. Therefore $T(n) = \Theta(g(n)) = \Theta(n^{1.585})$.

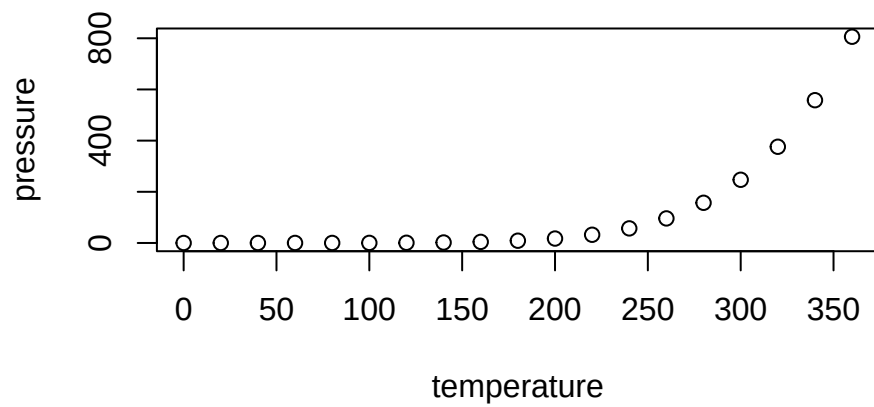
Answers to incidental LaTeX question may be found at:

<http://www.tug.org/begin.html>

`summary(cars)`

```
##      speed      dist
## Min.   : 4.0    Min.   : 2.00
## 1st Qu.:12.0    1st Qu.: 26.00
## Median :15.0    Median : 36.00
## Mean   :15.4    Mean   : 42.98
## 3rd Qu.:19.0    3rd Qu.: 56.00
## Max.   :25.0    Max.   :120.00
```

```
# command
plot(pressure)
```



NEW PAGE!!!